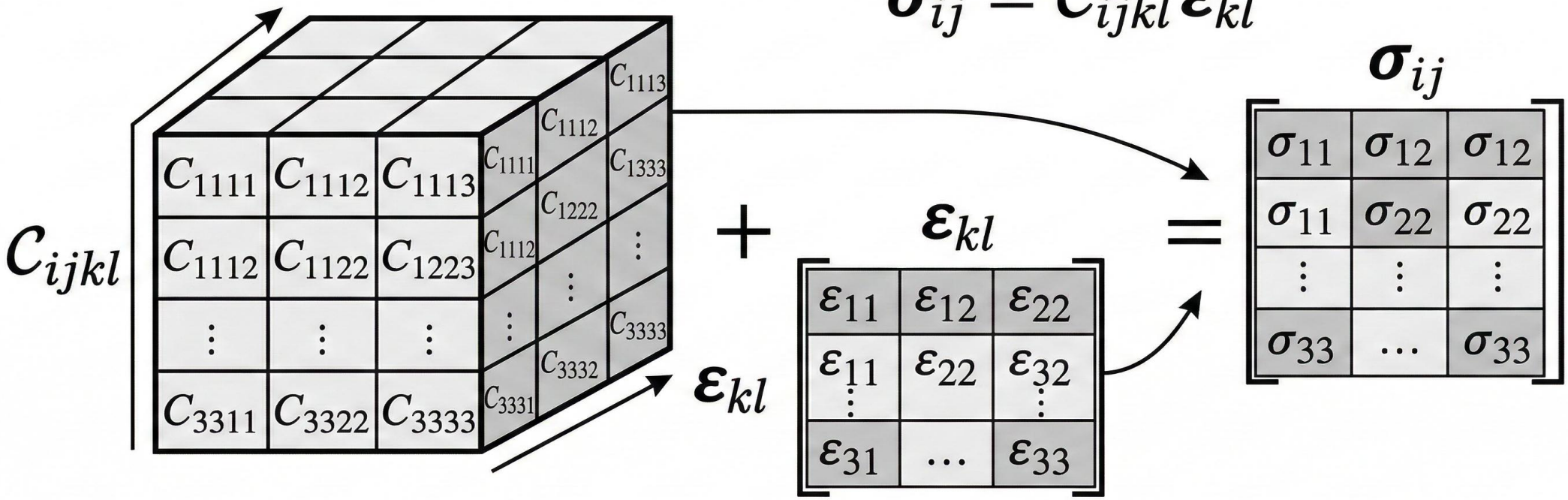
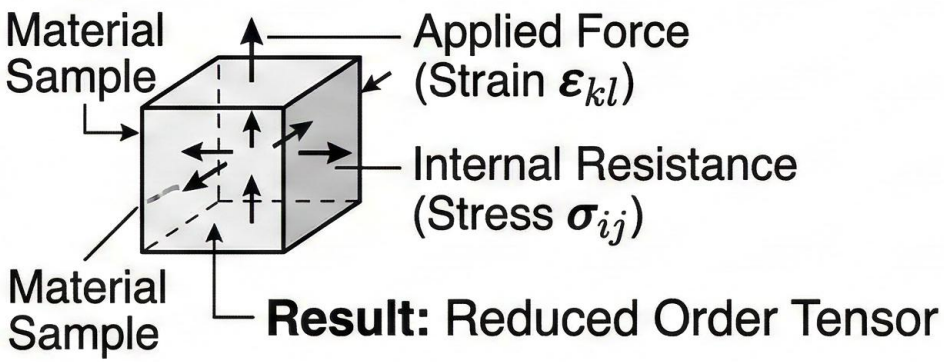
1. Hooke's Law: Contraction of Tensors

$$\boldsymbol{\sigma}_{ij} = C_{ijkl}\boldsymbol{\varepsilon}_{kl}$$



Physical Context



2. Visualization of Contraction Logic

$$C_{ijkl}\boldsymbol{\varepsilon}_{kl} \longrightarrow \sum_{k=1}^{3}\sum_{l=1}^{3} C_{ijkl}\boldsymbol{\varepsilon}_{kl}$$

$$\sigma_{11} = C_{1111}\varepsilon_{11} + C_{1112}\varepsilon_{12} + \ldots + C_{1133}\varepsilon_{33}$$